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**College of Science and Technology**
**Rinchending: Bhutan**

# WEB101
# Web Application Fundamentals
# (SS2026)

# Practical 1 Report

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**RUB Wheel of Academic Law: Academic Dishonesty**

**Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:**

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorised group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

**Aim:**

Cloning a website of TikTok.

## Theory:

The clone of a web application, say TikTok, is not the actual application. The clone is built to mimic the functionality of the existing application. The clone will allow users to watch videos, give likes, add comments, and even share their videos. The idea behind developing this type of clone is to learn the basics of web development and get a clear idea about APIs and database management.

The clone of the popular video-sharing application TikTok will include the following:
The frontend will include frameworks such as React or Lynx, along with Tailwind CSS, HTML, JavaScript, and CSS, used for designing the application.

The backend, or the server-side, will include Node.js and Express.js, used for user authentication, video uploading, and user interactions.

MongoDB will be used as the database, used to store all the relevant data.

## Background:

TikTok is a popular video-sharing application that was founded in 2016 and has revolutionized the world of social media. The application has changed the way users consume and interact with social media platforms. Unlike other popular video-sharing applications such as Facebook, Instagram, and YouTube, which allow users to connect and interact with each other, TikTok is different and focuses more on the content than the users. The application has been able to achieve this and has been widely popular. The popularity of the application has encouraged other popular video-sharing applications to add this feature to their platforms. For instance, applications such as Instagram and YouTube have been forced to add this feature to their platforms to remain at par with the popularity of TikTok. The popularity of the application has also encouraged developers to develop similar applications to understand how such applications can be created. A TikTok clone is created to help developers learn how to develop such applications.

## Implementation Steps

Implementation of Steps:
STEP 1: Open Vs-Code terminal and initialize a new Next.js project with the following command:

- npx create-next-app@latest and enter your project name*

When prompted, configure your project with these settings:
Would you like to use TypeScript? → No
Would you like to use ESLint? → Yes
Would you like to use Tailwind CSS? → Yes
Would you like to use src/ directory? → Yes
Would you like to use App Router? → Yes
Would you like to customize the default import alias? → No
STEP 2: Navigate to the src → app→ page.js
Replace the contents of page.js with this code:

```
import VideoFeed from "@/components/ui/VideoFeed";

export default function Home() {
  return (
    <div className="py-4">
      <VideoFeed/>
    </div>
  );
}
```

STEP 3: Navigate to src → app → global.css
Clean up globals.css and keep only the Tailwind directives:

- @import "tailwindcss".
- Install Tailwind with Next.js

STEP 4:
Create these folders (inside the src directory):

- mkdir -p src/components/layout
- mkdir -p src/components/ui
- mkdir -p src/lib
- mkdir -p src/app/profile
- mkdir -p src/app/upload
-

STEP 5: Create a basic layout component in:
Src → components → layout → MainLayout.jsx and put this code:

```
'use client';
import Link from "next/link";
import{
    FaHome, FaUserFriends, FaCompass, FaVideo,
    FaInbox, FaRegUser, FaPlus
} from 'react-icons/fa';

export default function MainLayout({ children }) {
    return (
        <div className="flex min-h-screen">
            {/* Sidebar */}
            <div className="w-60 border-r fixed h-full overflow-y-auto">
                <div className="p-4">
                    <Link href="/" className="text-xl font-bold flex items-center">
                        <span className="text-red-500 mr-1">Tiktok</span>
                    </Link>
                </div>

                <nav className="mt-4">
                    <ul className="space-y-2">
                        <li>
                            <Link
                            href="/"
                            className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                            >
                                <FaHome className="text-xl mr-3" />
                                <span>For You</span>
                            </Link>
                        </li>
                        <li>
                            <Link
                            href="/following"
                            className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                            >
                                <FaUserFriends className="text-xl mr-3" />
                                <span>Following</span>
                            </Link>
                        </li>
                        <li>
```

```
                <Link
                href="/explore"
                className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                >
                    <FaCompass className="text-xl mr-3" />
                    <span>Explore</span>
                </Link>
            </li>
            <li>
                <Link
                href="/live"
                className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                >
                    <FaVideo className="text-xl mr-3" />
                    <span>LIVE</span>
                </Link>
            </li>
        </ul>
    </nav>

    <div className="border-t mt-4 pt-4 px-2">
        <p className="text-gray-500 text-sm px-3 mb-2">Suggested accounts</p>
        {Array.from({ length: 5}).map ((_, index) => (
            <div key = {index} className="flex items-centger p-2 hoer:bg-gray-100 rounded-md">
                <div className="h-8 w-8 rounded-full bg-gray-300 mr-2"></div>
                <div>
                    <p className="text-sm font-semibold">user_{index + 1}</p>
                    <p className="text-xs text-gray-500">User {index + 1}</p>
                </div>
            </div>
        ))}
    </div>

    <div className="px-3 py-4 mt-2">
        <p className="text-sm text-gray-500 mb-4">
            Log in to follow creators, like videos, and view comments.
        </p>
```

```
        <Link href="/login">
            <button className="w-full py-2 px-4 border rounded-md font-medium mb-2 hover:bg-gray-50">
                Log in
            </button>
        </Link>

        <Link href="/signup">
            <button className="w-full py-2 px-4 bg-red-500 text-white rounded-md font-medium hover:bg-red-600">
            Sign up
            </button>
        </Link>
    </div>

    <div className="border-t px-3 py-4 text-xs text-gray-500">
        <p className="mb-2"> © 2025 Tiktok</p>
    </div>
/div>

/* Main content */}
div className="ml-60 flex-1">
    <div className="max-w-[1150px] mx-auto">
        {/* top header with search */}
        <header className="h-16 border-b flex items-center justify-between px-4">
            <div className="w-1/3"></div>
            <div className="w-1/3">
                <div className="relative">
                    <input
                        type="text"
                        placeholder="Seacrh accounts and videos"
                        className="w-full bg-gray-100 py-2 pl-10 pr-4 rounded-full"
                    />
                    <FaCompass className="absoulute left-4 top-1/2 transform -translate-y-1/2 text-gray-400" />
                </div>
            </div>
            <div className="w-1/3 flex justify-end space-x-4">
                <Link href="/upload">
                    <button className="border px-3 py-1 rounded-md hover:bg-gray-50 flex items-center">
                        <FaPlus className="mr-2" /> Upload
                    </button>
                </Link>
```

```
                    <button className="bg-red-500 text-white px-6 py-1 rounded-md">
                        Log in
                    </button>
                </div>
                <div>
                    <Link
                    href="/profile"
                    className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                    >
                        <FaRegUser className="text-xl mr-3" />
                        <span>Profile</span>
                    </Link>
                </div>
            </header>

            {/* Main content */}
            <main>{children}</main>
        </div>
    </div>
</div>
);
}
```

STEP 6: Update layout.js in the App Directory
Modify src → app → layout.js and code this:

```
import { Inter } from "next/font/google";
import "./globals.css";
import MainLayout from '@/components/layout/MainLayout';

const inter = Inter({ subsets: ['latin'] });

export const metadata = {
  title: "Tiktok Clone",
  description: 'A TikTok clone built with Next.js',
};

export default function RootLayout({ children }) {
  return (
    <html lang="en">
      <body className="{inter.className">
        <MainLayout>
          {children}
        </MainLayout>
      </body>
    </html>
  );
}
```

STEP 7: Create a profile page at:

- src → app → profile → page.jsx using this code

```jsx
import { FaEdit, FaUserPlus } from 'react-icons/fa';

export default function ProfilePage() {
  return (
    <div className="max-w-4xl mx-auto py-8">

      {/* Profile header */}
      <div className="flex items-start mb-8">
        <div className="h-24 w-24 rounded-full bg-gray-300 mr-6"></div>

        <div className="flex-1">
          <h1 className="text-xl font-bold mb-2">@username</h1>
          <h2 className="text-lg mb-4">User Name</h2>

          <button className="px-6 py-1.5 rounded-md border border-gray-300 font-medium flex items-center mr-3">
            <FaEdit className="mr-2" /> Edit profile
          </button>

          <div className="flex items-center mt-4 space-x-4">
            <p><span className="font-bold">0</span> Following</p>
            <p><span className="font-bold">0</span> Followers</p>
            <p><span className="font-bold">0</span> Likes</p>
          </div>
```

```jsx
            <p className="mt-4 text-sm max-w-md">
              No bio yet.
            </p>
          </div>
        </div>

        {/* Video tabs */}
        <div className="border-b">
          <div className="flex">
            <button className="py-2 px-4 border-b-2 border-black font-medium">
              Videos
            </button>
            <button className="py-2 px-4 text-gray-500 font-medium">
              Liked
            </button>
          </div>
        </div>

        {/* Videos grid */}
        <div className="py-4">
          <div className="grid grid-cols-6 gap-1">

            {/* Empty state for now */}
            <div className="col-span-6 py-20 text-center">
              <div className="max-w-md mx-auto">
                <h3 className="text-xl font-bold mb-2">Upload your first video</h3>
                <p className="text-gray-500 mb-6">Your videos will appear here</p>
                <button className="bg-red-500 text-white px-8 py-2 rounded-md font-medium">
                  Upload now
                </button>
              </div>
            </div>

          </div>
        </div>

      </div>
    );
  }
```

STEP 8: Creating an upload page at:

- src → app → upload → page.jsx → use this code:

```
export default function UploadPage() {
  return (
    <div className="max-w-4xl mx-auto p-8">
      <h1 className="text-2xl font-bold mb-6">Upload video</h1>

      <div className="flex">
        <div className="[360px] border-dashed border-2 border-gray-300 rounded-lg p-8 flex flex-col items-center justify-center text-center">
          <div className="w-16 h-16 rounded-full bg-gray-100 flex items-center justify-center mb-4">
            <span className="text-4xl text-gray-400">+</span>
          </div>
          <h3 className="text-lg font-semibold mb-2">Select video to upload</h3>
          <p className="text-sm text-gray-500 mb-4">Or drag and drop a file</p>
          <p className="text-xs text-gray-400 mb-4">MP4 or WebM</p>
          <p className="text-xs text-gray-400 mb-4">720x1280 resolution or higher</p>
          <p className="text-xs text-gray-400 mb-4">Up to 10 minutes</p>
          <p className="text-xs text-gray-400">Less than 2 GB</p>
          <button className="mt-8 bg-red-500 text-white py-2 px-8 rounded-md">
            Select file
          </button>
        </div>

        <div className="flex-1 ml-6">
          <div className="mb-4">
            <label className="block text-sm font-medium mb-2">Caption</label>
            <input
              type="text"
              className="w-full p-2 border rounded-md"
              placeholder="Add a caption..."
            />
          </div>

          <div className="mb-4">
            <label className="block text-sm font-medium mb-2">Cover</label>
            <div className="h-20 bg-gray-200 rounded-md"></div>
          </div>

          <div className="mb-4">
            <label className="block text-sm font-medium mb-2">
              Who can view this video
```

```
            </label>
            <select className="w-full p-2 border rounded-md">
              <option>Public</option>
              <option>Friends</option>
              <option>Private</option>
            </select>
          </div>

          <div className="mb-4">
            <label className="block text-sm font-medium mb-2">
              Allow users to:
            </label>
            <div className="space-y-2">
              <div className="flex items-center">
                <input type="checkbox" id="comments" className="mr-2" />
                <label htmlFor="comments">Comment</label>
              </div>
              <div className="flex items-center">
                <input type="checkbox" id="duet" className="mr-2" />
                <label htmlFor="duet">Duet</label>
              </div>
              <div className="flex items-center">
                <input type="checkbox" id="stitch" className="mr-2" />
                <label htmlFor="stitch">Stitch</label>
              </div>
            </div>
          </div>

          <div className="flex space-x-3 mt-6">
            <button className="px-6 py-2 border rounded-md">Discard</button>
            <button className="px-6 py-2 bg-red-500 text-white rounded-md">
              Post
            </button>
          </div>
        </div>
      </div>
    </div>
  );
}
```

STEP 9: Start the Development Server:

- In VS-code terminal type: install npm → cd practical1 → npm run dev
- Run at: http://localhost:3000

Part 2: Creating the Web Layout and Main Interface

Step 1: Installing Additional Dependencies

Open Vs-code terminal and type:

- npm install react-icons

Step 2: Update the Layout structure

Go to src → components → layout → MainLayout.jsx:

```
'use client';
import Link from "next/link";
import{
    FaHome, FaUserFriends, FaCompass, FaVideo,
    FaInbox, FaRegUser, FaPlus
} from 'react-icons/fa';

export default function MainLayout({ children }) {
    return (
        <div className="flex min-h-screen">
            {/* Sidebar */}
            <div className="w-60 border-r fixed h-full overflow-y-auto">
                <div className="p-4">
                    <Link href="/" className="text-xl font-bold flex items-center">
                        <span className="text-red-500 mr-1">Tiktok</span>
                    </Link>
                </div>

                <nav className="mt-4">
                    <ul className="space-y-2">
                        <li>
                            <Link
                            href="/"
                            className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                            >
                                <FaHome className="text-xl mr-3" />
                                <span>For You</span>
                            </Link>
                        </li>
                        <li>
                            <Link
                            href="/following"
                            className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                            >
                                <FaUserFriends className="text-xl mr-3" />
                                <span>Following</span>
                            </Link>
                        </li>
                        <li>
```

```
                    <Link
                    href="/explore"
                    className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                    >
                        <FaCompass className="text-xl mr-3" />
                        <span>Explore</span>
                    </Link>
                </li>
                <li>
                    <Link
                    href="/live"
                    className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                    >
                        <FaVideo className="text-xl mr-3" />
                        <span>LIVE</span>
                    </Link>
                </li>
            </ul>
        </nav>

        <div className="border-t mt-4 pt-4 px-2">
            <p className="text-gray-500 text-sm px-3 mb-2">Suggested accounts</p>
            {Array.from({ length: 5}).map ((_, index) => (
                <div key = {index} className="flex items-centger p-2 hoer:bg-gray-100 rounded-md">
                    <div className="h-8 w-8 rounded-full bg-gray-300 mr-2"></div>
                    <div>
                        <p className="text-sm font-semibold">user_{index + 1}</p>
                        <p className="text-xs text-gray-500">User {index + 1}</p>
                    </div>
                </div>
            ))}
        </div>

        <div className="px-3 py-4 mt-2">
            <p className="text-sm text-gray-500 mb-4">
                Log in to follow creators, like videos, and view comments.
            </p>
```

```
        <Link href="/login">
            <button className="w-full py-2 px-4 border rounded-md font-medium mb-2 hover:bg-gray-50">
                Log in
            </button>
        </Link>

        <Link href="/signup">
            <button className="w-full py-2 px-4 bg-red-500 text-white rounded-md font-medium hover:bg-red-600">
            Sign up
            </button>
        </Link>
    </div>

    <div className="border-t px-3 py-4 text-xs text-gray-500">
        <p className="mb-2"> © 2025 Tiktok</p>
    </div>
/div>

/* Main content */}
div className="ml-60 flex-1">
    <div className="max-w-[1150px] mx-auto">
        {/* top header with search */}
        <header className="h-16 border-b flex items-center justify-between px-4">
            <div className="w-1/3"></div>
            <div className="w-1/3">
                <div className="relative">
                    <input
                        type="text"
                        placeholder="Seacrh accounts and videos"
                        className="w-full bg-gray-100 py-2 pl-10 pr-4 rounded-full"
                    />
                    <FaCompass className="absoulute left-4 top-1/2 transform -translate-y-1/2 text-gray-400" />
                </div>
            </div>
            <div className="w-1/3 flex justify-end space-x-4">
                <Link href="/upload">
                    <button className="border px-3 py-1 rounded-md hover:bg-gray-50 flex items-center">
                        <FaPlus className="mr-2" /> Upload
                    </button>
                </Link>
```

```
                    <button className="bg-red-500 text-white px-6 py-1 rounded-md">
                        Log in
                    </button>
                </div>
                <div>
                    <Link
                    href="/profile"
                    className="flex items-center p-3 hover:bg-gray-100 rounded-md mx-2"
                    >
                        <FaRegUser className="text-xl mr-3" />
                        <span>Profile</span>
                    </Link>
                </div>
            </header>

            {/* Main content */}
            <main>{children}</main>
        </div>
    </div>
</div>
);
}
```

STEP 3: Create a For You Page like that of TikTok's web version.
Create a new file at: src → components → ui → VideoCard.jsx:

```
'use client';
import {useState } from 'react';
import {
    FaHeart, FaRegHeart, FaComment,
    FaShare, FaMusic
} from 'react-icons/fa';

export default function VideoCard ({post}) {
    const [liked, setLiked] = useState(false);

    //Placeholder data
    const { username, caption, audio, likes, comments, shares } = post;
    const handleLikeClick = () => {
        setLiked(!liked);
    };

    return (
        <div className="flex py-6 border-b">
            {/* User avatar */}
            <div className="mr-3">
                <div className="h-12 w-12 rounded-full bg-gray-300"></div>
            </div>

            <div className="flex-1">
                {/* User info and caption */}
                <div className="mb-2">
                    <span className="text-sm ml-1"> 2d ago </span>
                    <p className="text-sm mt-1"> {caption} </p>
                </div>

                {/* Audio info */}
                <div className="flex items-center text-sm mb-3">
                    <FaMusic className="mr-2 text-xs" />
                    <span className="truncate max-w-[250px]">{audio}</span>
                </div>
```

```
                <div className="flex">
                    {/* Video conatiner */}
                    <div className="mr-5 w-[300px] h-[530px] bg-black rounded-md flex items-center justify-center relative overflow
                        <p className="text-white"> Video Placeholder</p>
                        <div className="absolute bottom-4 left-4 text-white text-sm">
                            <p className="mb-1"> 0:30 </p>
                        </div>
                    </div>
                    {/* Interaction buttons */}
                    <div className="flex flex-col justify-end space-y-3 py-2">
                        {/* Like button */}
                        <button
                        className="flex flex-col items-center"
                        onClick = {handleLikeClick}
                        >
                            <div className="h-10 w-10 rounded-full bg-gray-200 flex items-center justidy-center">
                                {liked ? (
                                    <FaHeart className="text-red-500" />
                                ) : (
                                    <FaRegHeart />
                                )}
                            </div>
                            <span className="text-xs mt-1"> {liked ? likes + 1 : likes}</span>
                        </button>

                        {/* Comment button */}
                        <button className="flex flex-col items-center">
                            <div className="h-10 w-10 rounded-full bg-gray-200 flex items-center justify-center">
                                <FaShare />
                            </div>
                            <span className="text-xs mt-1"> {shares}</span>
                        </button>
                    </div>
                </div>
            </div>
        </div>
    );
```

STEP 4: Similarly Create a VideoFeed Component

- Create a new file at src → components → ui → VideoFeed.jsx and code this:

```
'use client';
import VideoCard from "./VideoCard";

// Sample data for our feed
const DUMMY_POSTS = [
    {
        id: '1',
        username: '@user1',
        caption: 'Check out this cool video! #trending #tikotk #viral',
        audio: 'original Sounds - User1',
        likes: 1234,
        comments: 432,
        shares: 89
    },
    {
        id: '2',
        username: '@user2',
        caption: 'Learning to dance #dance #fun #trending',
        audio: 'Popular Song - Artist',
        likes: 5678,
        comments: 321,
        shares: 52
    },
    {
        id: '3',
        username: '@user3',
        caption: 'Beautiful sunset today! #Nature #Sunset #Vibes',
        audio: 'Sunset vibes - Chilll Music',
        likes: 2468,
        comments: 135,
        shares: 46
    }
];

export default function VideoFeed () {
    return (
        <div className="max-w-[550px] mx-auto">
            {DUMMY_POSTS.map((post) => (
                <VideoCard key= {post.id} post={post} />
            ))}
        </div>
    );
}
```

STEP 5: Update the Home Page to display the video feed.
Go to src → app → page.js:

```
import VideoFeed from "@/components/ui/VideoFeed";

export default function Home() {
  return (
    <div className="py-4">
      <VideoFeed/>
    </div>
  );
}
```

STEP 6: Create a new Following Page at:
Src → app → following → page.jsx:

```
export default function FollowingPage() {
  return (
    <div className="max-w-[550px] mx-auto py-10 text-center">
      <h2 className="text-2xl font-bold mb-3">Follow accounts</h2>
      <p className="text-gray-500 mb-8">
        Follow accounts to see their latest videos
      </p>

      <div className="grid grid-cols-5 gap-4 mb-6">
        {Array.from({ length: 5 }).map((_, index) => (
          <div key={index} className="text-center">
            <div className="h-16 w-16 rounded-full bg-gray-300 mx-auto mb-2"></div>
            <p className="text-sm font-semibold">user_{index + 1}</p>
            <button className="mt-2 text-xs bg-red-500 text-white py-1 px-4 rounded-full">
              Follow
            </button>
          </div>
        ))}
      </div>

      <button className="border border-gray-300 text-gray-700 py-2 px-8 rounded-md font-medium">
        See more
      </button>
    </div>
  );
}
```

STEP 7: Create a new Explore file at:
Src → app → explore →page.jsx:

```
export default function ExplorePage() {
    return (
      <div className="p-4">
        <h2 className="text-2xl font-bold mb-4">Explore</h2>

        <div className="mb-8">
          <h3 className="text-xl font-semibold mb-3">Trending Hashtags</h3>
          <div className="grid grid-cols-3 gap-4">
            {Array.from({ length: 6 }).map((_, index) => (
              <div
                key={index}
                className="aspect-video bg-gray-200 rounded-md flex flex-col items-center justify-center p-4"
              >
                <p className="font-bold text-lg">#Trending{index + 1}</p>
                <p className="text-sm text-gray-500">{(index + 1) * 1.5}M views</p>
              </div>
            ))}
          </div>
        </div>

        <div>
          <h3 className="text-xl font-semibold mb-3">Popular Videos</h3>
          <div className="grid grid-cols-4 gap-3">
            {Array.from({ length: 12 }).map((_, index) => (
              <div
                key={index}
                className="aspect-[9/16] bg-gray-300 rounded-md flex items-center justify-center"
              >
                <p className="text-sm">Video {index + 1}</p>
              </div>
            ))}
          </div>
        </div>
      </div>
    );
  }
```

STEP 8: Create a new Live Page file at:
Src → app → live → page.jsx:

```jsx
export default function LivePage() {
  return (
    <div className="p-4">
      <h2 className="text-2xl font-bold mb-4">LIVE</h2>

      <div className="grid grid-cols-3 gap-4">
        {Array.from({ length: 9 }).map((_, index) => (
          <div key={index} className="rounded-md overflow-hidden">
            <div className="aspect-video bg-gray-200 flex items-center justify-center">
              <p>LIVE Stream {index + 1}</p>
            </div>

            <div className="p-2 border border-t-0 rounded-b-md">
              <div className="flex items-center">
                <div className="h-8 w-8 rounded-full bg-gray-300 mr-2"></div>

                <di
                  <
                  <p className="text-xs text-gray-500">Live stream title here</p>
                </div>
              </div>

              <div className="mt-2 text-xs text-red-500">
                {(index + 1) * 125} viewers
              </div>
            </div>
          </div>
        ))}
      </div>
    </div>
  );
}
```

(property) JSX.IntrinsicElements.p: DetailedHTMLProps<HTMLAttributes<HTMLParagraphElement>, HTMLParagraphElement>

STEP 9: update upload page to match TikTok's web interface by clicking:

- Src → app → upload → page.jsx:

```jsx
export default function UploadPage() {
  return (
    <div className="max-w-4xl mx-auto p-8">
      <h1 className="text-2xl font-bold mb-6">Upload video</h1>

      <div className="flex">
        <div className="[360px] border-dashed border-2 border-gray-300 rounded-lg p-8 flex flex-col items-center justify-center text-center">
          <div className="w-16 h-16 rounded-full bg-gray-100 flex items-center justify-center mb-4">
            <span className="text-4xl text-gray-400">+</span>
          </div>
          <h3 className="text-lg font-semibold mb-2">Select video to upload</h3>
          <p className="text-sm text-gray-500 mb-4">Or drag and drop a file</p>
          <p className="text-xs text-gray-400 mb-4">MP4 or WebM</p>
          <p className="text-xs text-gray-400 mb-4">720x1280 resolution or higher</p>
          <p className="text-xs text-gray-400 mb-4">Up to 10 minutes</p>
          <p className="text-xs text-gray-400">Less than 2 GB</p>
          <button className="mt-8 bg-red-500 text-white py-2 px-8 rounded-md">
            Select file
          </button>
        </div>

        <div className="flex-1 ml-6">
          <div className="mb-4">
            <label className="block text-sm font-medium mb-2">Caption</label>
            <input
              type="text"
              className="w-full p-2 border rounded-md"
              placeholder="Add a caption..."
            />
          </div>

          <div className="mb-4">
            <label className="block text-sm font-medium mb-2">Cover</label>
            <div className="h-20 bg-gray-200 rounded-md"></div>
          </div>
```

```
<div className="mb-4">
  <label className="block text-sm font-medium mb-2">
    Who can view this video
  </label>
  <select className="w-full p-2 border rounded-md">
    <option>Public</option>
    <option>Friends</option>
    <option>Private</option>
  </select>
</div>

<div className="mb-4">
  <label className="block text-sm font-medium mb-2">
    Allow users to:
  </label>
  <div className="space-y-2">
    <div className="flex items-center">
      <input type="checkbox" id="comments" className="mr-2" />
      <label htmlFor="comments">Comment</label>
    </div>
    <div className="flex item
      <input type="checkbox" id="duet" className="mr-2" />
      <label htmlFor="duet">Duet</label>
    </div>
    <div className="flex items-center">
      <input type="checkbox" id="stitch" className="mr-2" />
      <label htmlFor="stitch">Stitch</label>
    </div>
  </div>
</div>

<div className="flex space-x-3 mt-6">
  <button className="px-6 py-2 border rounded-md">Discard</button>
  <button className="px-6 py-2 bg-red-500 text-white rounded-md">
    Post
  </button>
</div>
</div>
</div>
</div>
);
}
```

(property) HTMLAttributes<HTMLInputElement>.id?: string

STEP 10: Update the Profile Page

- Go to src → app → profile → page.jsx:

```jsx
import { FaEdit, FaUserPlus } from 'react-icons/fa';

export default function ProfilePage() {
  return (
    <div className="max-w-4xl mx-auto py-8">

      {/* Profile header */}
      <div className="flex items-start mb-8">
        <div className="h-24 w-24 rounded-full bg-gray-300 mr-6"></div>

        <div className="flex-1">
          <h1 className="text-xl font-bold mb-2">@username</h1>
          <h2 className="text-lg mb-4">User Name</h2>

          <button className="px-6 py-1.5 rounded-md border border-gray-300 font-medium flex items-center mr-3">
            <FaEdit className="mr-2" /> Edit profile
          </button>

          <div className="flex items-center mt-4 space-x-4">
            <p><span className="font-bold">0</span> Following</p>
            <p><span className="font-bold">0</span> Followers</p>
            <p><span className="font-bold">0</span> Likes</p>
          </div>

          <p className="mt-4 text-sm max-w-md">
            No bio yet.
          </p>
        </div>
      </div>
```

```jsx
      {/* Video tabs */}
      <div className="border-b">
        <div className="flex">
          <button className="py-2 px-4 border-b-2 border-black font-medium">
            Videos
          </button>
          <button className="py-2 px-4 text-gray-500 font-medium">
            Liked
          </button>
        </div>
      </div>

      {/* Videos grid */}
      <div className="py-4">
        <div className="grid grid-cols-6 gap-1">

          {/* Empty state for now */}
          <div className="col-span-6 py-20 text-center">
            <div className="max-w-md mx-auto">
              <h3 className="text-xl font-bold mb-2">Upload your first video</h3>
              <p className="text-gray-500 mb-6">Your videos will appear here</p>
              <button className="bg-red-500 text-white px-8 py-2 rounded-md font-medium">
                Upload now
              </button>
            </div>
          </div>

        </div>
      </div>

    </div>
  );
}
```

STEP 11: Test Your Application:

- Run your development server and test the application by clicking npm run dev in terminal

Part 3: Creating Login and Registration Forms for TikTok

Objectives:

● Create a login page
● Create a registration page
● Implement form validation
● Connect forms to the navigation in MainLayout.jsx
● Understand form handling concepts

STEP11.1: install a package to help with form handling and validation:

Type in terminal:

- npm install react-hook-form

STEP 11.2: Create a new Login Page file at:

- src → app → login →page.jsx:

```
'use client';

import { useState } from 'react';
import { useForm } from 'react-hook-form';
import Link from 'next/link';

export default function LoginPage() {
  const [isLoading, setIsLoading] = useState(false);

  const {
    register,
    handleSubmit,
    formState: { errors },
  } = useForm();

  const onSubmit = async (data) => {
    setIsLoading(true);

    // In a real app, you would call an authentication API here
    console.log('Login data:', data);

    // Simulate API call
    setTimeout(() => {
      setIsLoading(false);
      alert('Login successful (demo only)');
    }, 1500);
  };

  return (
    <div className="flex flex-col items-center justify-center py-8 px-4">
      <div className="w-full max-w-md">

        <div className="text-center mb-8">
          <h1 className="text-2xl font-bold">Log in to TikTok</h1>
          <p className="text-gray-500 mt-2">
            Manage your account, check notifications, comment on videos, and more
          </p>
        </div>

        <div className="border rounded-lg p-6">
          <form onSubmit={handleSubmit(onSubmit)}>

            <div className="mb-4">
              <input
```

```
    type="text"
    placeholder="Email or username"
    className="w-full px-3 py-2 border rounded-md focus:outline-none focus:ring-1 focus:ring-gray-400"
    {...register('email', {
      required: 'Email or username is required',
    })}
  />
  {errors.email && (
    <p className="text-red-500 text-xs mt-1">
      {errors.email.message}
    </p>
  )}
</div>

<div className="mb-4">
  <input
    type="password"
    placeholder="Password"
    className="w-full px-3 py-2 border rounded-md focus:outline-none focus:ring-1 focus:ring-gray-400"
    {...register('password', {
      required: 'Password is required',
    })}
  />
  {errors.password && (
    <p className="text-red-500 text-xs mt-1">
      {errors.password.message}
    </p>
  )}
</div>

<div className="mb-4 text-right">
  <Link
    href="/reset-password"
    className="text-sm text-gray-500 hover:underline"
  >
    Forgot password?
  </Link>
</div>
```

```
            <button
              type="submit"
              className="w-full bg-red-500 text-white py-3 rounded-md font-medium hover:bg-red-600 transition"
              disabled={isLoading}
            >
              {isLoading ? 'Logging in...' : 'Log in'}
            </button>

          </form>
        </div>

        <div className="mt-4 text-center">
          <p className="text-gray-500">
            Don't have an account?{' '}
            <Link
              href="/signup"
              className="text-red-500 font-medium hover:underline"
            >
              Sign up
            </Link>
          </p>
        </div>

      </div>
    </div>
  );
}
```

STEP 11.3: Create a new Signup Page file at:

- src → app → signup → page.jsx:

```jsx
'use client';

import { useState } from 'react';
import { useForm } from 'react-hook-form';
import Link from 'next/link';

export default function SignupPage() {
  const [isLoading, setIsLoading] = useState(false);
  const { register, handleSubmit, watch, formState: { errors } } = useForm();

  const onSubmit = async (data) => {
    setIsLoading(true);

    // In a real app, you would call a registration API here
    console.log('Signup data:', data);

    // Simulate API call
    setTimeout(() => {
      setIsLoading(false);
      alert('Registration successful (demo only)');
    }, 1500);
  };

  const password = watch('password');

  return (
    <div className="min-h-screen flex items-center justify-center bg-gray-50 py-12 px-4 sm:px-6 lg:px-8">
      <div className="max-w-md w-full space-y-8">

        <div className="text-center">
          <h2 className="text-3xl font-extrabold text-gray-900">
            Sign up for TikTok
          </h2>
          <p className="mt-2 text-sm text-gray-600">
            Create a profile, follow other accounts, make your own videos, and more
          </p>
        </div>

        <form className="mt-8 space-y-6" onSubmit={handleSubmit(onSubmit)}>
          <div className="rounded-md shadow-sm -space-y-px">
```

```
<div className="mb-4">
  <label htmlFor="username" className="sr-only">Username</label>
  <input
    id="username"
    type="text"
    className="appearance-none rounded-none relative block w-full px-3 py-2 border border-gray-300 placeholder-gray-500 text-gray-900 r
    placeholder="Username"
    {...register('username', {
      required: 'Username is required',
      minLength: {
        value: 3,
        message: 'Username must be at least 3 characters'
      },
      pattern: {
        value: /^[a-zA-Z0-9_]+$/,
        message: 'Username can only contain letters, numbers and underscores'
      }
    })}
  />
  {errors.username && (
    <p className="text-red-500 text-xs mt-1">{errors.username.message}</p>
  )}
</div>

{/* Email */}
<div className="mb-4">
  <label htmlFor="email" className="sr-only">Email address</label>
  <input
    id="email"
    type="email"
    className="appearance-none rounded-none relative block w-full px-3 py-2 border border-gray-300 placeholder-gray-500 text-gray-900 focu
    placeholder="Email address"
    {...register('email', {
      required: 'Email is required',
      pattern: {
        value: /^[A-Z0-9._%+-]+@[A-Z0-9.-]+\.[A-Z]{2,}$/i,
        message: 'Invalid email address'
      }
    })}
  />
  {errors.email && (
    <p className="text-red-500 text-xs mt-1">{errors.email.message}</p>
  )}
```

```
</div>

{/* Password */}
<div className="mb-4">
  <label htmlFor="password" className="sr-only">Password</label>
  <input
    id="password"
    type="password"
    className="appearance-none rounded-none relative block w-full px-3 py-2 border border-gray-300 placeholder-gray-500 text-gray-900 f
    placeholder="Password"
    {...register('password', {
      required: 'Password is required',
      minLength: {
        value: 8,
        message: 'Password must be at least 8 characters'
      },
      pattern: {
        value: /^(?=.[a-z])(?=.[A-Z])(?=.\d)(?=.[$@$!%?&])[A-Za-z\d$@$!%?&]{8,}$/,
        message:
          'Password must contain at least one uppercase letter, one lowercase letter, one number and one special character'
      }
    })}
  />
  {errors.password && (
    <p className="text-red-500 text-xs mt-1">{errors.password.message}</p>
  )}
</div>

{/* Confirm Password */}
<div>
  <label htmlFor="confirmPassword" className="sr-only">
    Confirm Password
  </label>
  <input
    id="confirmPassword"
    type="password"
    className="appearance-none rounded-none relative block w-full px-3 py-2 border border-gray-300 placeholder-gray-500 text-gray-900 r
    placeholder="Confirm Password"
    {...register('confirmPassword', {
      required: 'Please confirm your password',
      validate: value =>
```

```
            value === password || 'Passwords do not match'
          })}
        />
        {errors.confirmPassword && (
          <p className="text-red-500 text-xs mt-1">
            {errors.confirmPassword.message}
          </p>
        )}
      </div>

    </div>

    {/* Terms */}
    <div className="flex items-center">
      <input
        id="terms"
        type="checkbox"
        className="h-4 w-4 text-red-600 focus:ring-red-500 border-gray-300 rounded"
        {...register('terms', {
          required: 'You must agree to the terms and conditions'
        })}
      />
      <label htmlFor="terms" className="ml-2 block text-sm text-gray-900">
        I agree to the{' '}
        <a href="#" className="font-medium text-red-600 hover:text-red-500">
          Terms of Service
        </a>{' '}
        and{' '}
        <a href="#" className="font-medium text-red-600 hover:text-red-500">
          Privacy Policy
        </a>
      </label>
    </div>

    {errors.terms && (
      <p className="text-red-500 text-xs">{errors.terms.message}</p>
    )}
```

```
            {/* Button */}
            <div>
              <button
                type="submit"
                className="group relative w-full flex justify-center py-2 px-4 border border-transparent text-sm font-medium rounded-md text-white b
                disabled={isLoading}
              >
                {isLoading ? 'Creating account...' : 'Sign up'}
              </button>
            </div>
          </form>

          {/* Login Link */}
          <div className="text-center mt-4">
            <p className="text-sm text-gray-600">
              Already have an account?{' '}
              <Link
                href="/login"
                className="font-medium text-red-600 hover:text-red-500"
              >
                Log in
              </Link>
            </p>
          </div>

        </div>
      </div>
    );
  }
```

STEP 11.4: Update MainLayout Component to include login and signup by tapping:

Change 1: Update the Login Button in the Sidebar

- src → components → layout →MainLayout.jsx and replace the login and sign-up button by using this code:

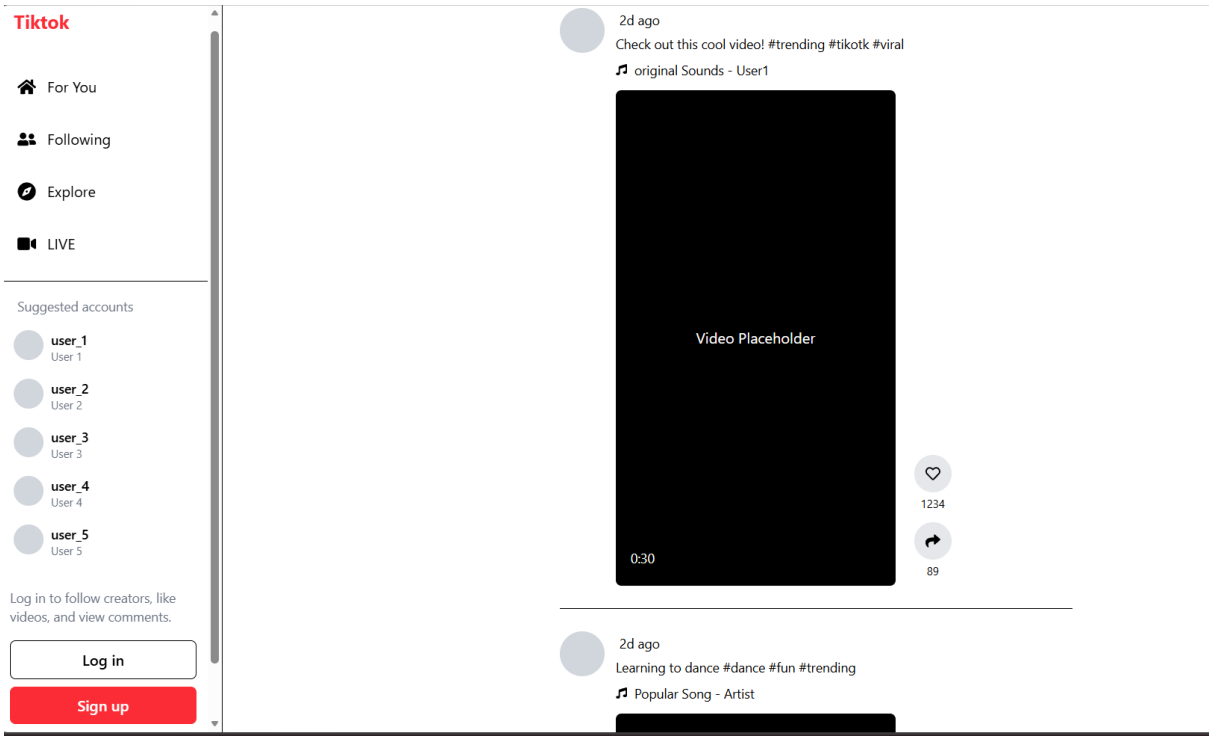
```
<div className="px-3 py-4 mt-2">
    <p className="text-sm text-gray-500 mb-4">
        Log in to follow creators, like videos, and view comments.
    </p>

    <Link href="/login">
        <button className="w-full py-2 px-4 border rounded-md font-medium mb-2 hover:bg-gray-50">
            Log in
        </button>
    </Link>

    <Link href="/signup">
        <button className="w-full py-2 px-4 bg-red-500 text-white rounded-md font-medium hover:bg-red-600">
        Sign up
        </button>
    </Link>
</div>
```

*Change 2: Update the Header Login Button*

- *Similarly go to src → components → layout →MainLayout.jsx and replace the code of login head button by using this code:*

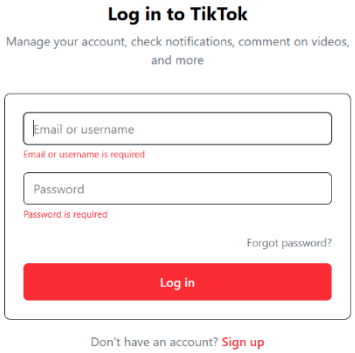
```
<Link href="/login">
  <button className="bg-red-500 text-white px-6 py-1 rounded-md hover:bg-red-600">
    Log in
  </button>
</Link>
```

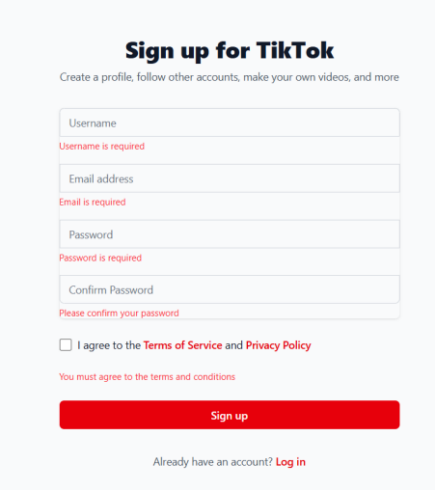
Final Result:



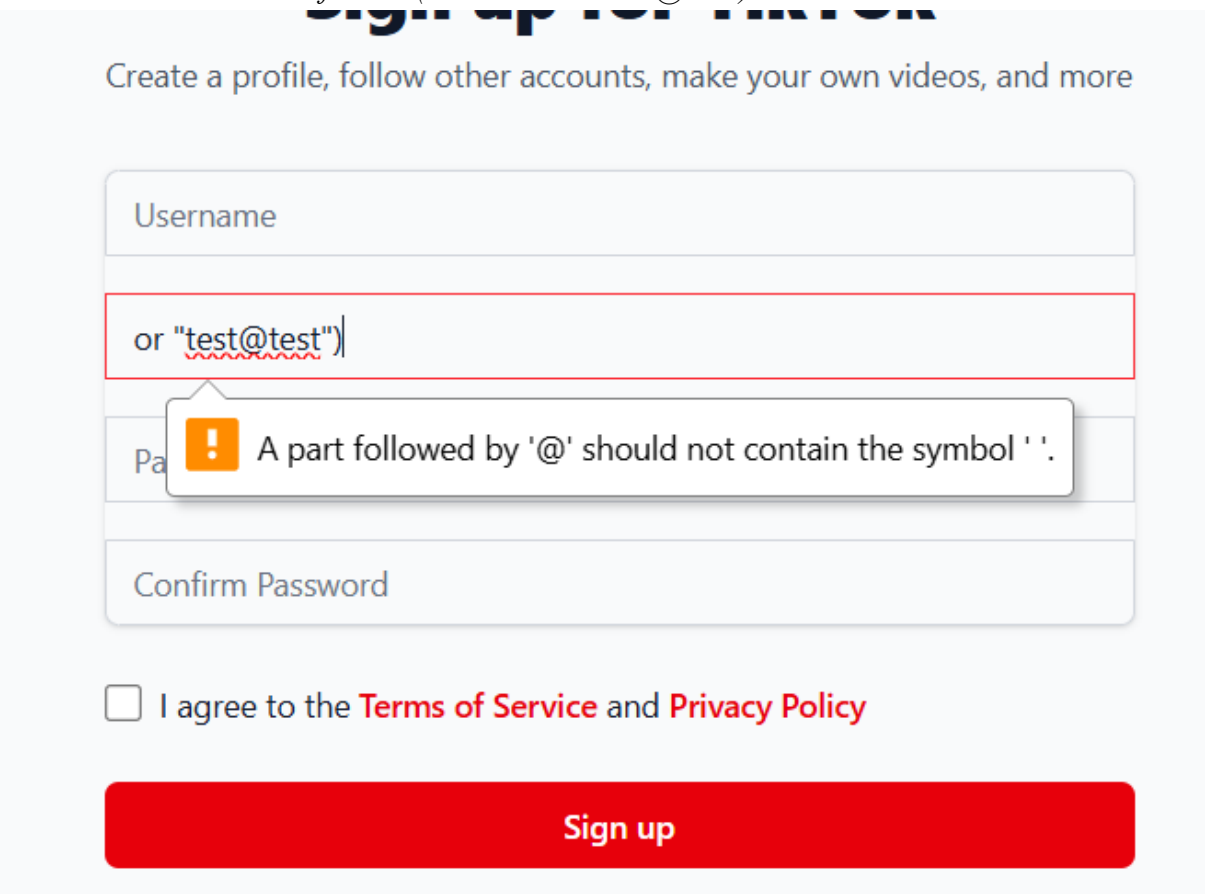
**Testing the Login and Signup Forms Test form validation:**

- Try submitting the forms without filling in any fields

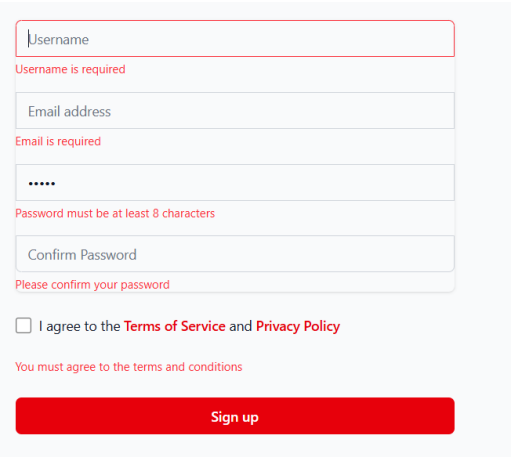


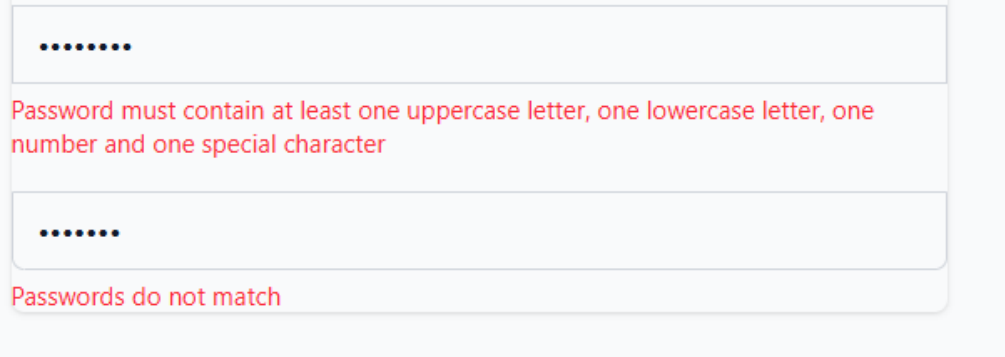

- *Enter an invalid email format (like "test" or "test@test")*



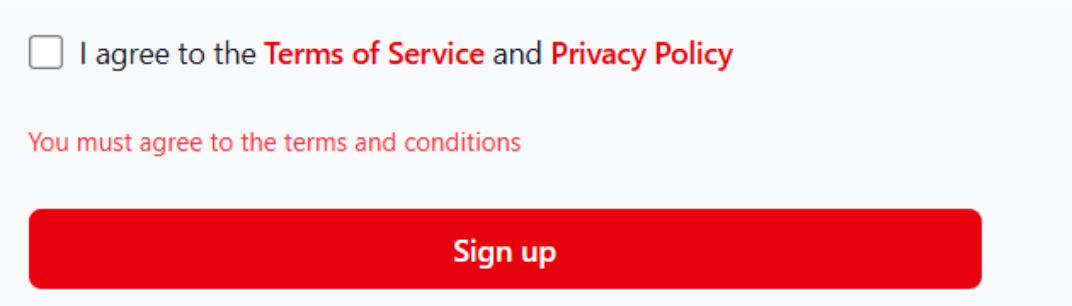

- *Enter a short password (less than 8 characters)*



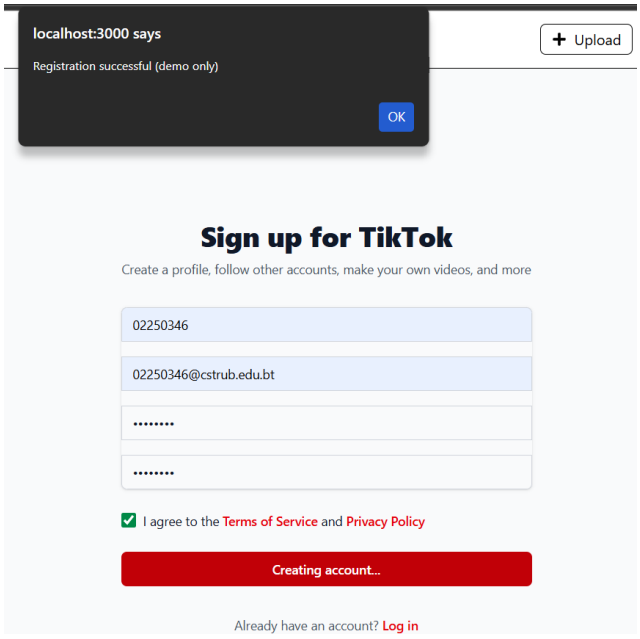

● *For signup, try passwords that don't match in both fields*




● *Attempt to submit the signup form without checking the terms checkbox Test successful submission:*



- *Fill in all fields correctly and submit*
- *Verify that the loading state appears*
- *Confirm you see the success message*




## Conclusion

In this practical exercise, a TikTok-like web application was created using existing templates and adapting relevant code structures to mimic the main features of the application. This project was used to demonstrate how web applications are created using existing technology and frameworks.

The process of building this web application involved creating a user interface that mimics the original application and incorporating main features such as video playing, interactive features such as liking, and user authentication features such as logging in and registering on the application.

During this project, relevant concepts were learned and implemented, including component-based architecture, form validation techniques, and how to manage user interactions and components. This practical also helped in gaining more knowledge about front-end development and provided hands-on experience in building web applications.

## Reference


- *React.js: https://legacy.reactjs.org/docs*
- *Next.js: https://nextjs.org/docs*
- *Linking and Navigating: https://nextjs.org/docs/app/building-your-application/routing/linking-and-navigating*
- *React components and props: https://legacy.reactjs.org/docs/components-and-props.html*
- *React Hook Forms: https://www.freecodecamp.org/news/how-to-create-forms-in-react-using-react-hook-form/*
- *Formik: https://formik.org/ Regular Expressions (Regex): https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Regular_expressions*
- *Validating React Hook Forms (with zod): https://youtu.be/cc_xmawJ8Kg?si=0MxIQZGmbexQf-73*